OpenCV Matrix Assignment Report

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1. Objective

This report demonstrates that a digital image is a numerical matrix rather than only a visual object. An image captured on a mobile phone is converted to 200 x 200 pixels, its pixel values are exported to CSV matrices, several OpenCV operations are applied, and selected operations are verified against manual matrix calculations.

2. Image Acquisition and Preparation

The source photograph was captured with a mobile phone and stored as `input/image_original.jpg`, then resized to 200 x 200 pixels with `cv2.resize` using `INTER_AREA` interpolation and saved as `input/image_200x200.png`.

Image Properties

The properties below were read from the final 200 x 200 image with OpenCV and are stored in `csv_full_image/image_metadata.csv`.

property	value
width	200
height	200
channels	3
shape	(200, 200, 3)
dtype	uint8
channel_order	BGR
min_pixel_value	1
max_pixel_value	227
mean_pixel_value	149.1377
std_pixel_value	63.6612
gray_min_pixel_value	2
gray_max_pixel_value	224
gray_mean_pixel_value	150.2941
gray_std_pixel_value	63.6493

Channel order. `cv2.imread` loads a color image in **BGR order**, not RGB order. Index 0 of the third axis is blue, index 1 is green, and index 2 is red. All channel matrices in this report follow that convention, and `cv2.split` returns the channels in the same B, G, R sequence.

3. The Image as a Numerical Matrix

Each channel of the 200 x 200 image is exported in full to `csv_full_image/` as a 200 x 200 grid of integers in the range 0-255. Each file contains numerical values only, with no row names, column headers, or DataFrame index. Because a full grid is too large to print, the top-left 8 x 8 corner of each channel is shown below; the complete values are in the CSV files.

Grayscale (`image_gray_200x200.csv`)

	c0	c1	c2	c3	c4	c5	c6	c7
r0	199	200	200	200	200	200	200	200
r1	199	199	199	200	199	199	200	201

	c0	c1	c2	c3	c4	c5	c6	c7
r2	199	200	200	200	199	200	200	201
r3	200	200	199	200	199	200	200	201
r4	199	200	200	199	199	200	199	200
r5	200	199	199	200	200	199	199	200
r6	200	200	199	199	200	199	200	200
r7	199	200	198	200	200	200	199	200

Blue channel (`image_blue_200x200.csv`)

	c0	c1	c2	c3	c4	c5	c6	c7
r0	194	195	195	195	195	195	196	196
r1	194	194	194	195	194	194	196	197
r2	194	195	195	195	194	195	195	196
r3	195	195	194	195	194	195	195	196
r4	194	195	195	194	194	195	194	195
r5	195	194	194	195	195	194	194	195
r6	194	194	194	194	195	194	195	195
r7	193	194	193	195	195	195	194	195

Green channel (`image_green_200x200.csv`)

	c0	c1	c2	c3	c4	c5	c6	c7
r0	201	202	202	202	202	201	202	201
r1	201	201	201	202	201	201	202	202
r2	201	202	202	202	201	201	201	202
r3	202	202	201	202	201	202	202	203
r4	201	202	202	201	201	202	201	202
r5	202	201	201	202	202	201	201	202
r6	202	202	201	201	202	201	202	202
r7	201	201	200	202	202	202	201	202

Red channel (`image_red_200x200.csv`)

	c0	c1	c2	c3	c4	c5	c6	c7
r0	198	199	199	199	199	199	199	199
r1	198	198	198	199	198	198	199	200
r2	198	199	199	199	198	199	199	200
r3	199	199	198	199	198	199	199	200
r4	198	199	199	198	198	199	198	199
r5	199	198	198	199	199	198	198	199
r6	199	199	198	198	199	198	199	199
r7	198	199	197	199	199	199	198	199

Grayscale Calculation

Grayscale conversion is a weighted sum of the three color channels rather than a plain average. The weights reflect how sensitive human vision is to each color: the eye is most sensitive to green, less to red, and least to blue. OpenCV uses

`I_gray = round(0.114 * B + 0.587 * G + 0.299 * R)`

with the channels taken in BGR order as loaded by `cv2.imread`. The weights sum to 1.0, so the result stays within the original 0-255 range.

The equation was applied by hand to 5 pixels taken from different locations in the image and compared against the value OpenCV produced.

row	column	B	G	R	manual_gray	opencv_gray	difference
0	0	194	201	198	199	199	0
50	150	84	83	92	86	86	0
100	100	79	93	146	107	107	0
150	50	114	130	133	129	129	0
199	199	147	160	162	159	159	0

Every manually calculated value matches OpenCV exactly, confirming the weighted equation is the operation OpenCV performs.

4. OpenCV Operations

Each operation below was applied to the prepared image. Every visual result is saved in `output_images/` as a PNG and its numerical output is saved in `csv_full_image/` as a CSV matrix. Unless stated otherwise, operations are performed on the grayscale image.

4.1 Color and Intensity Operations

Grayscale

Weighted BGR conversion to a single intensity channel.

Top-left 8 x 8 corner of `grayscale_matrix.csv`:

	c0	c1	c2	c3	c4	c5	c6	c7
r0	199	200	200	200	200	200	200	200
r1	199	199	199	200	199	199	200	201
r2	199	200	200	200	199	200	200	201
r3	200	200	199	200	199	200	200	201
r4	199	200	200	199	199	200	199	200
r5	200	199	199	200	200	199	199	200
r6	200	200	199	199	200	199	200	200
r7	199	200	198	200	200	200	199	200

Channel Blue

Blue channel isolated with `cv2.split`, shown as intensities.

Channel Green

Green channel isolated with `cv2.split`.

Channel Red

Red channel isolated with `cv2.split`.

Merged Color

The three channels merged back with `cv2.merge([B, G, R])`. The result is verified to be identical to the original image, which confirms the BGR order.

Negative

`I_negative = 255 − I`, inverting every intensity.

Top-left 8 x 8 corner of `negative_matrix.csv`:

	c0	c1	c2	c3	c4	c5	c6	c7
r0	56	55	55	55	55	55	55	55
r1	56	56	56	55	56	56	55	54
r2	56	55	55	55	56	55	55	54
r3	55	55	56	55	56	55	55	54
r4	56	55	55	56	56	55	56	55
r5	55	56	56	55	55	56	56	55
r6	55	55	56	56	55	56	55	55
r7	56	55	57	55	55	55	56	55

Brightness Plus40

`I_bright = clip(I + 40, 0, 255)`. Clipping prevents values above 255 from wrapping around to small numbers.

Top-left 8 x 8 corner of `brightness_plus40_matrix.csv`:

	c0	c1	c2	c3	c4	c5	c6	c7
r0	239	240	240	240	240	240	240	240
r1	239	239	239	240	239	239	240	241
r2	239	240	240	240	239	240	240	241
r3	240	240	239	240	239	240	240	241
r4	239	240	240	239	239	240	239	240
r5	240	239	239	240	240	239	239	240
r6	240	240	239	239	240	239	240	240
r7	239	240	238	240	240	240	239	240

Contrast 1 25

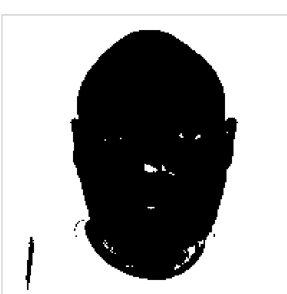
`I_contrast = clip(1.25 x I, 0, 255)`. Values spread away from the midpoint, so bright areas saturate at 255.

Top-left 8 x 8 corner of `contrast_1_25_matrix.csv`:

	c0	c1	c2	c3	c4	c5	c6	c7
r0	249	250	250	250	250	250	250	250
r1	249	249	249	250	249	249	250	251
r2	249	250	250	250	249	250	250	251
r3	250	250	249	250	249	250	250	251
r4	249	250	250	249	249	250	249	250
r5	250	249	249	250	250	249	249	250
r6	250	250	249	249	250	249	250	250
r7	249	250	248	250	250	250	249	250

Threshold

`I_binary = 255` where `I > 127`, otherwise `0`. Produces the binary image used for the morphological and contour sections.

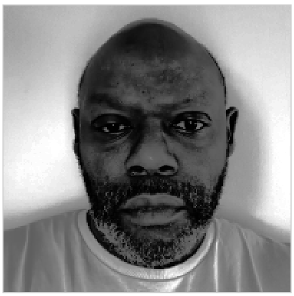


Top-left 8 x 8 corner of `threshold_matrix.csv`:

	c0	c1	c2	c3	c4	c5	c6	c7
r0	255	255	255	255	255	255	255	255
r1	255	255	255	255	255	255	255	255
r2	255	255	255	255	255	255	255	255
r3	255	255	255	255	255	255	255	255
r4	255	255	255	255	255	255	255	255
r5	255	255	255	255	255	255	255	255
r6	255	255	255	255	255	255	255	255
r7	255	255	255	255	255	255	255	255

Histogram Equalized

Histogram equalization redistributes intensities so the cumulative histogram is approximately linear, increasing global contrast.

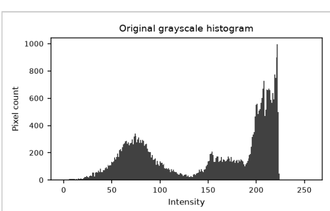


Top-left 8 x 8 corner of `histogram_equalized_matrix.csv`:

	c0	c1	c2	c3	c4	c5	c6	c7
r0	158	162	162	162	162	162	162	162
r1	158	158	158	162	158	158	162	165
r2	158	162	162	162	158	162	162	165
r3	162	162	158	162	158	162	162	165
r4	158	162	162	158	158	162	158	162
r5	162	158	158	162	162	158	158	162
r6	162	162	158	158	162	158	162	162
r7	158	162	155	162	162	162	158	162

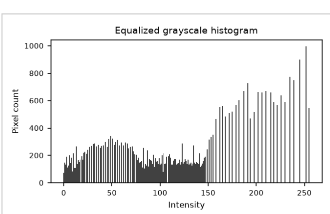
Histogram Original

Intensity distribution before equalization.



Histogram Equalized Plot

Intensity distribution after equalization.



4.2 Geometric Operations

Center Crop 100X100

The center 100 x 100 region, rows and columns 50 to 149.

Top-left 8 x 8 corner of `center_crop_100x100_matrix.csv`:

	c0	c1	c2	c3	c4	c5	c6	c7
r0	206	205	203	180	75	64	73	75
r1	206	205	200	145	63	66	73	75
r2	205	204	196	114	58	66	74	75
r3	205	203	193	81	61	69	74	74
r4	204	202	171	66	67	73	73	73
r5	204	198	141	57	70	74	72	73
r6	205	193	120	57	76	75	74	74
r7	205	197	119	65	75	75	75	75

Flip Horizontal

`cv2.flip` with code 1 reverses column order.

Top-left 8 x 8 corner of `flip_horizontal_matrix.csv`:

	c0	c1	c2	c3	c4	c5	c6	c7
r0	211	211	211	212	211	210	211	210
r1	210	211	211	211	211	210	211	211
r2	211	211	211	211	211	210	211	211
r3	212	210	211	212	210	211	212	211
r4	211	210	211	211	211	212	212	211
r5	211	212	212	211	211	211	211	211
r6	212	212	212	211	211	211	212	211
r7	212	211	212	212	211	211	211	211

Flip Vertical

`cv2.flip` with code 0 reverses row order.

Top-left 8 x 8 corner of `flip_vertical_matrix.csv`:

	c0	c1	c2	c3	c4	c5	c6	c7
r0	164	161	158	155	152	152	156	163

	c0	c1	c2	c3	c4	c5	c6	c7
r1	164	161	159	154	151	153	156	163
r2	165	161	158	153	150	151	156	163
r3	164	160	157	152	149	149	154	163
r4	163	159	155	151	149	150	156	163
r5	164	158	154	150	148	149	154	162
r6	164	159	154	149	147	147	153	160
r7	164	159	154	148	147	146	150	157

Rotate 90

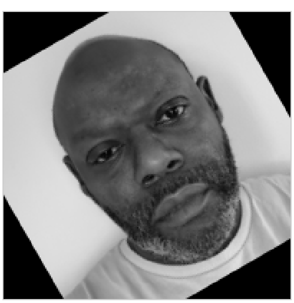
`cv2.rotate` by 90 degrees clockwise, a pure index transposition.

Top-left 8 x 8 corner of `rotate_90_matrix.csv`:

	c0	c1	c2	c3	c4	c5	c6	c7
r0	164	164	165	164	163	164	164	164
r1	161	161	161	160	159	158	159	159
r2	158	159	158	157	155	154	154	154
r3	155	154	153	152	151	150	149	148
r4	152	151	150	149	149	148	147	147
r5	152	153	151	149	150	149	147	146
r6	156	156	156	154	156	154	153	150
r7	163	163	163	163	163	162	160	157

Rotate 30

`cv2.warpAffine` with a 30 degree rotation about the center. Corners rotate outside the frame and are cropped, and empty corners are filled with black.



Top-left 8 x 8 corner of `rotate_30_matrix.csv`:

	c0	c1	c2	c3	c4	c5	c6	c7
r0	0	0	0	0	0	0	0	0
r1	0	0	0	0	0	0	0	0
r2	0	0	0	0	0	0	0	0
r3	0	0	0	0	0	0	0	0
r4	0	0	0	0	0	0	0	0
r5	0	0	0	0	0	0	0	0
r6	0	0	0	0	0	0	0	0
r7	0	0	0	0	0	0	0	0

Resized 100X100

Downsampled to 100 x 100 with `INTER_AREA`.

Top-left 8 x 8 corner of `resized_100x100_matrix.csv`:

	c0	c1	c2	c3	c4	c5	c6	c7
r0	199	200	200	200	201	201	201	202
r1	200	200	200	201	201	201	202	202
r2	200	200	200	200	201	201	201	202
r3	200	199	200	200	200	201	202	203
r4	200	199	200	200	200	201	201	202
r5	199	199	200	200	201	201	201	202
r6	199	199	200	200	201	201	202	202
r7	199	200	200	200	200	201	201	201

Upscaled Nearest

The 100 x 100 image returned to 200 x 200 with nearest neighbor.

Top-left 8 x 8 corner of `upscaled_nearest_matrix.csv`:

	c0	c1	c2	c3	c4	c5	c6	c7
r0	199	199	200	200	200	200	200	200
r1	199	199	200	200	200	200	200	200
r2	200	200	200	200	200	200	201	201
r3	200	200	200	200	200	200	201	201
r4	200	200	200	200	200	200	200	200
r5	200	200	200	200	200	200	200	200
r6	200	200	199	199	200	200	200	200
r7	200	200	199	199	200	200	200	200

Upscaled Bilinear

The 100 x 100 image returned to 200 x 200 with bilinear.

Top-left 8 x 8 corner of `upscaled_bilinear_matrix.csv`:

	c0	c1	c2	c3	c4	c5	c6	c7
r0	199	199	200	200	200	200	200	200
r1	199	199	200	200	200	200	200	200
r2	200	200	200	200	200	200	201	201

	c0	c1	c2	c3	c4	c5	c6	c7
r3	200	200	200	200	200	200	201	201
r4	200	200	200	200	200	200	200	200
r5	200	200	200	200	200	200	200	200
r6	200	200	199	199	200	200	200	200
r7	200	200	199	199	200	200	200	200

Nearest Neighbor Compared With Bilinear

Both upscaled images start from the same 100 x 100 downsample, so any difference comes purely from how each method invents the missing pixels.

Nearest neighbor copies the value of the closest source pixel. It performs no arithmetic, so every output value already existed in the source and edges stay hard. The cost is blockiness: each source pixel becomes a visible 2 x 2 square of identical values, which gives diagonal edges a stair-stepped appearance.

Bilinear interpolation takes a weighted average of the four nearest source pixels. This introduces intermediate values that were never in the source, producing smooth gradients and removing the blocky squares, at the cost of softening genuine edges. The result looks less sharp but closer to the original continuous image.

Measured against the original 200 x 200 grayscale image:

method	mean_abs_difference	max_abs_difference	std_of_result	unique_values
nearest_neighbor	3.1936	88	63.2336	213
bilinear	2.9863	72	62.6373	213

The bilinear result is closer to the original on both mean and maximum absolute difference, and its lower standard deviation reflects the smoothing it applies.

4.3 Spatial Filtering Operations

Filter Mean 3X3

3 x 3 mean filter using `K_mean = (1/9) * ones(3, 3)`.

Top-left 8 x 8 corner of `filter_mean_3x3_matrix.csv`:

	c0	c1	c2	c3	c4	c5	c6	c7
r0	199	199	200	200	200	200	200	201
r1	199	199	200	200	200	200	200	200
r2	200	199	200	199	200	200	200	201
r3	200	200	200	199	200	200	200	200
r4	200	200	200	199	200	199	200	200
r5	200	200	199	199	199	199	200	200
r6	200	199	199	199	200	200	200	200
r7	200	199	199	199	200	200	200	200

Filter Gaussian 3X3

3 x 3 Gaussian filter using `K_Gaussian = (1/16) * [[1,2,1],[2,4,2],[1,2,1]]`. Center-weighted, so it smooths less aggressively than the mean filter.

Top-left 8 x 8 corner of `filter_gaussian_3x3_matrix.csv`:

	c0	c1	c2	c3	c4	c5	c6	c7
r0	199	199	200	200	200	200	200	200
r1	199	199	200	200	200	200	200	201
r2	200	200	200	200	199	200	200	201
r3	200	200	200	200	199	200	200	200
r4	200	200	200	199	199	200	200	200
r5	200	200	199	200	200	199	199	200
r6	200	200	199	199	200	200	200	200
r7	200	200	199	199	200	200	200	200

Filter Median 3X3

3 x 3 median filter. Being rank-based rather than a convolution, it removes isolated outliers while keeping edges sharp.

Top-left 8 x 8 corner of `filter_median_3x3_matrix.csv`:

	c0	c1	c2	c3	c4	c5	c6	c7
r0	199	199	200	200	200	200	200	200
r1	199	199	200	200	200	200	200	200
r2	199	199	200	199	200	200	200	201
r3	200	200	200	199	200	200	200	200
r4	200	200	200	199	200	199	200	200
r5	200	200	199	199	199	199	200	200
r6	200	199	199	200	200	200	200	200
r7	200	200	199	199	200	200	200	200

4.4 Edge-Detection Operations

Sobel X

Horizontal Sobel using `G_x = [[-1,0,1],[-2,0,2],[-1,0,1]]`.

Top-left 8 x 8 corner of `sobel_x_matrix.csv`:

	c0	c1	c2	c3	c4	c5	c6	c7
r0	0	2	2	0	-2	2	4	4
r1	0	2	2	-1	-2	3	5	3
r2	0	1	1	-2	-1	4	5	2
r3	0	0	-1	-2	1	3	3	4
r4	0	0	-1	-1	1	0	2	7
r5	0	-2	0	2	-1	-2	3	6
r6	0	-4	-1	5	-1	-2	3	3
r7	0	-4	-2	6	1	-3	1	4

Sobel Y

Vertical Sobel using `G_y = [[-1,-2,-1],[0,0,0],[1,2,1]]`.

Top-left 8 x 8 corner of `sobel_y_matrix.csv`:

	c0	c1	c2	c3	c4	c5	c6	c7
r0	0	0	0	0	0	0	0	0
r1	0	0	0	-1	-2	-1	1	1
r2	4	3	1	0	1	2	1	0
r3	0	0	-1	-2	-1	-1	-3	-2
r4	-2	-2	-1	1	1	-2	-4	-3
r5	2	0	-2	0	1	0	1	0
r6	0	0	-1	-1	1	2	1	-1
r7	0	0	0	0	1	1	-1	0

Sobel Magnitude

Gradient magnitude `G = sqrt(G_x^2 + G_y^2)`.

Top-left 8 x 8 corner of `sobel_magnitude_matrix.csv`:

	c0	c1	c2	c3	c4	c5	c6	c7
r0	0	2	2	0	2	2	4	4
r1	0	2	2	1.4142	2.8284	3.1623	5.099	3.1623
r2	4	3.1623	1.4142	2	1.4142	4.4721	5.099	2
r3	0	0	1.4142	2.8284	1.4142	3.1623	4.2426	4.4721
r4	2	2	1.4142	1.4142	1.4142	2	4.4721	7.6158
r5	2	2	2	2	1.4142	2	3.1623	6
r6	0	4	1.4142	5.099	1.4142	2.8284	3.1623	3.1623
r7	0	4	2	6	1.4142	3.1623	1.4142	4

Laplacian

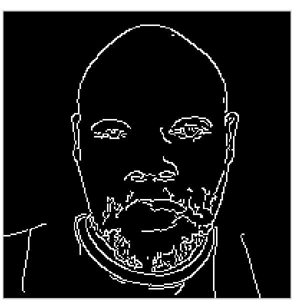
Second-derivative operator responding to intensity peaks and troughs.

Top-left 8 x 8 corner of `laplacian_matrix.csv`:

	c0	c1	c2	c3	c4	c5	c6	c7
r0	2	-3	-2	0	-2	-2	0	3
r1	0	2	3	-2	2	3	0	-2
r2	3	-2	-2	-1	2	-2	1	-2
r3	-2	-1	4	-3	2	-1	0	-2
r4	4	-2	-3	3	2	-3	3	1
r5	-3	3	2	-3	-2	2	2	0
r6	-1	-2	0	3	-2	3	-3	0
r7	4	-3	6	-4	0	-2	3	-1

Canny

Canny edge detection with gradient thresholds of 100 and 200.



Top-left 8 x 8 corner of `canny_matrix.csv`:

	c0	c1	c2	c3	c4	c5	c6	c7
r0	0	0	0	0	0	0	0	0
r1	0	0	0	0	0	0	0	0
r2	0	0	0	0	0	0	0	0
r3	0	0	0	0	0	0	0	0
r4	0	0	0	0	0	0	0	0
r5	0	0	0	0	0	0	0	0
r6	0	0	0	0	0	0	0	0
r7	0	0	0	0	0	0	0	0

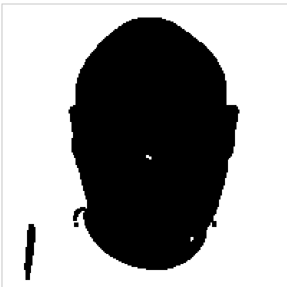
Data Type of Gradient Values

Sobel and Laplacian responses are signed: an edge running dark to light gives a positive value and the same edge running light to dark gives a negative one. Both operators are therefore computed with `cv2.CV_64F` rather than the default 8-bit unsigned type. Writing the result straight into a `uint8` matrix would clip every negative value to zero and silently discard half of the detected edges. The CSV matrices in `csv_full_image/` hold the true signed values; the PNG previews are separately scaled to the 0-255 display range.

4.5 Morphological Operations

Morph Erosion

Erosion with a 3 x 3 kernel of ones. A pixel stays white only when every pixel under the kernel is white, so white regions shrink.

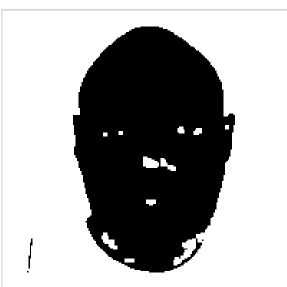


Top-left 8 x 8 corner of `morph_erosion_matrix.csv`:

	c0	c1	c2	c3	c4	c5	c6	c7
r0	255	255	255	255	255	255	255	255
r1	255	255	255	255	255	255	255	255
r2	255	255	255	255	255	255	255	255
r3	255	255	255	255	255	255	255	255
r4	255	255	255	255	255	255	255	255
r5	255	255	255	255	255	255	255	255
r6	255	255	255	255	255	255	255	255
r7	255	255	255	255	255	255	255	255

Morph Dilation

Dilation with the same kernel. A pixel becomes white when at least one pixel under the kernel is white, so white regions grow.

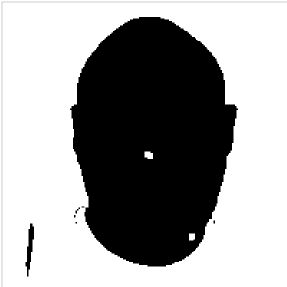


Top-left 8 x 8 corner of `morph_dilation_matrix.csv`:

	c0	c1	c2	c3	c4	c5	c6	c7
r0	255	255	255	255	255	255	255	255
r1	255	255	255	255	255	255	255	255
r2	255	255	255	255	255	255	255	255
r3	255	255	255	255	255	255	255	255
r4	255	255	255	255	255	255	255	255
r5	255	255	255	255	255	255	255	255
r6	255	255	255	255	255	255	255	255
r7	255	255	255	255	255	255	255	255

Morph Opening

Erosion followed by dilation, removing small white specks.

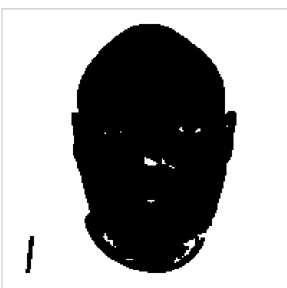


Top-left 8 x 8 corner of `morph_opening_matrix.csv`:

	c0	c1	c2	c3	c4	c5	c6	c7
r0	255	255	255	255	255	255	255	255
r1	255	255	255	255	255	255	255	255
r2	255	255	255	255	255	255	255	255
r3	255	255	255	255	255	255	255	255
r4	255	255	255	255	255	255	255	255
r5	255	255	255	255	255	255	255	255
r6	255	255	255	255	255	255	255	255
r7	255	255	255	255	255	255	255	255

Morph Closing

Dilation followed by erosion, filling small black holes.



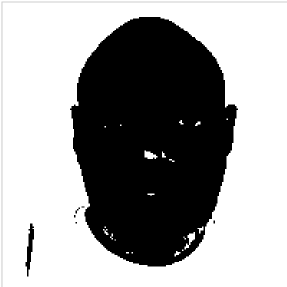
Top-left 8 x 8 corner of `morph_closing_matrix.csv`:

	c0	c1	c2	c3	c4	c5	c6	c7
r0	255	255	255	255	255	255	255	255
r1	255	255	255	255	255	255	255	255
r2	255	255	255	255	255	255	255	255
r3	255	255	255	255	255	255	255	255
r4	255	255	255	255	255	255	255	255
r5	255	255	255	255	255	255	255	255
r6	255	255	255	255	255	255	255	255
r7	255	255	255	255	255	255	255	255

4.6 Contour Analysis

Contour Mask

All detected contours filled white on a black background.

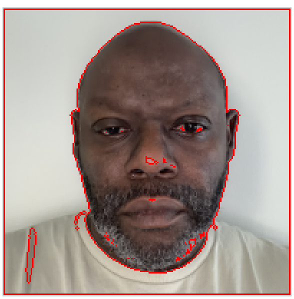


Top-left 8 x 8 corner of `contour_mask_matrix.csv` :

	c0	c1	c2	c3	c4	c5	c6	c7
r0	255	255	255	255	255	255	255	255
r1	255	255	255	255	255	255	255	255
r2	255	255	255	255	255	255	255	255
r3	255	255	255	255	255	255	255	255
r4	255	255	255	255	255	255	255	255
r5	255	255	255	255	255	255	255	255
r6	255	255	255	255	255	255	255	255
r7	255	255	255	255	255	255	255	255

Contours Drawn

Every detected contour outlined in red on the original image.



Contour Measurements

Contours were detected from the binary threshold image, which yielded 30 contours. `RETR_LIST` was used rather than `RETR_EXTERNAL` because the white region reaches every image border: with `RETR_EXTERNAL` the only contour returned traces the image frame itself and describes nothing about the image content. The frame contour is still present in the table below and is flagged in the `is_image_frame` column, but it is excluded when identifying the largest contour.

The full table is saved as `csv_full_image/contour_measurements.csv`. The ten largest contours are shown here.

contour_id	area	perimeter	bbox_x	bbox_y	bbox_width	bbox_height	centroid_x	centroid_y	points	is_image_frame	is_largest
29	39601	796	0	0	200	200	99.5	99.5	4	True	False
28	15435.5	505.002	46	9	119	176	103.597	95.8123	207	False	True
11	142.5	84.0416	15	153	8	39	18.5661	170.608	30	False	False
22	22	23.3137	99	103	9	6	102.212	106.091	14	False	False
7	18.5	31.5563	127	160	7	12	130.739	164.009	18	False	False
13	4	7.6569	146	151	3	4	147	152.5	6	False	False
14	4	7.6569	49	147	3	4	50	148.5	6	False	False
15	4	7.6569	50	144	3	4	51	145.5	6	False	False
18	4	7.6569	54	141	4	3	55.5	142	6	False	False
9	3.5	12.2426	136	156	4	5	137.81	157.476	8	False	False

Largest contour

- Contour id: 28
- Area: 15435.5
- Perimeter: 505.0021
- Bounding box x, y: 46, 9
- Bounding box width, height: 119, 176

- Centroid: (103.5968, 95.8123)

5. Manual Matrix Calculations

Manual calculation is performed on a single 7 x 7 grayscale patch rather than the full 200 x 200 image. Every manual result below was produced by explicit arithmetic on the pixel values: point operations were evaluated element by element, and neighborhood operations were evaluated by sliding the kernel by hand and summing the products. No manual result was produced by calling the OpenCV function it is compared against.

For every 3 x 3 neighborhood operation only the central 5 x 5 output is compared. Those 25 output pixels depend solely on values inside the patch, so OpenCV's border-padding rules cannot influence the comparison.

Selected Patch

The patch spans **rows 121.0 to 127.0** and **columns 53.0 to 59.0** of the 200 x 200 grayscale image. It was chosen as the 7 x 7 window with the highest standard deviation (83.867), so it contains noticeable intensity variation: values range from 18.0 to 213.0.

Saved as `csv_manual_calculations/manual_input_patch_7x7.csv`.

	c0	c1	c2	c3	c4	c5	c6
r0	212	207	151	38	30	21	41
r1	212	210	178	46	34	29	22
r2	213	211	185	83	24	22	18
r3	213	211	202	101	42	22	19
r4	213	210	204	151	30	18	22
r5	212	210	207	172	36	36	38
r6	211	210	207	192	87	36	36

Summary of All Manual Operations

operation_id	operation	output_shape	max_abs_difference	result
op01	Grayscale conversion (5 color pixels)	5x1	0	MATCH
op02	Negative transformation	7x7	0	MATCH
op03	Brightness adjustment (+40)	7x7	0	MATCH
op04	Contrast adjustment (x1.25)	7x7	0	MATCH
op05	Binary thresholding (>127)	7x7	0	MATCH
op06	Horizontal flip	7x7	0	MATCH
op07	Mean filter (3x3)	5x5	0	MATCH
op08	Gaussian filter (3x3)	5x5	0	MATCH
op09	Median filter (3x3)	5x5	0	MATCH
op10	Sobel Gx	5x5	0	MATCH
op11	Sobel Gy	5x5	0	MATCH
op12	Sobel gradient magnitude	5x5	0	MATCH
op13	Erosion (3x3 ones)	5x5	0	MATCH
op14	Dilation (3x3 ones)	5x5	0	MATCH

All 14 of 14 operations reproduce the OpenCV result. Where a maximum difference is not exactly zero it is on the order of 1e-14, which is floating-point representation error rather than a disagreement in the arithmetic.

OP01 - Grayscale conversion (5 color pixels)

Input (`op01_grayscale_input.csv`)

	c0	c1	c2	c3	c4
r0	0	0	194	201	198
r1	50	150	84	83	92
r2	100	100	79	93	146
r3	150	50	114	130	133
r4	199	199	147	160	162

Manual output (`op01_grayscale_manual_output.csv`)

	c0
r0	199
r1	86
r2	107
r3	129
r4	159

OpenCV output (`op01_grayscale_opencv_output.csv`)

	c0
r0	199
r1	86
r2	107
r3	129
r4	159

Difference (manual - OpenCV) (`op01_grayscale_difference.csv`)

	c0
r0	0
r1	0
r2	0
r3	0
r4	0

Maximum absolute difference: **0.0** (MATCH).

OP02 - Negative transformation

Input (`op02_negative_input.csv`)

	c0	c1	c2	c3	c4	c5	c6
r0	212	207	151	38	30	21	41
r1	212	210	178	46	34	29	22
r2	213	211	185	83	24	22	18
r3	213	211	202	101	42	22	19
r4	213	210	204	151	30	18	22
r5	212	210	207	172	36	36	38
r6	211	210	207	192	87	36	36

Manual output (`op02_negative_manual_output.csv`)

	c0	c1	c2	c3	c4	c5	c6
r0	43	48	104	217	225	234	214
r1	43	45	77	209	221	226	233
r2	42	44	70	172	231	233	237
r3	42	44	53	154	213	233	236
r4	42	45	51	104	225	237	233
r5	43	45	48	83	219	219	217
r6	44	45	48	63	168	219	219

OpenCV output (`op02_negative_opencv_output.csv`)

	c0	c1	c2	c3	c4	c5	c6
r0	43	48	104	217	225	234	214
r1	43	45	77	209	221	226	233
r2	42	44	70	172	231	233	237
r3	42	44	53	154	213	233	236
r4	42	45	51	104	225	237	233
r5	43	45	48	83	219	219	217
r6	44	45	48	63	168	219	219

Difference (manual - OpenCV) (`op02_negative_difference.csv`)

	c0	c1	c2	c3	c4	c5	c6
r0	0	0	0	0	0	0	0
r1	0	0	0	0	0	0	0
r2	0	0	0	0	0	0	0
r3	0	0	0	0	0	0	0
r4	0	0	0	0	0	0	0
r5	0	0	0	0	0	0	0
r6	0	0	0	0	0	0	0

Maximum absolute difference: **0.0** (MATCH).

OP03 - Brightness adjustment (+40)

Input (`op03_brightness_input.csv`)

	c0	c1	c2	c3	c4	c5	c6
r0	212	207	151	38	30	21	41
r1	212	210	178	46	34	29	22
r2	213	211	185	83	24	22	18
r3	213	211	202	101	42	22	19
r4	213	210	204	151	30	18	22
r5	212	210	207	172	36	36	38
r6	211	210	207	192	87	36	36

Manual output (`op03_brightness_manual_output.csv`)

	c0	c1	c2	c3	c4	c5	c6
r0	252	247	191	78	70	61	81
r1	252	250	218	86	74	69	62
r2	253	251	225	123	64	62	58
r3	253	251	242	141	82	62	59
r4	253	250	244	191	70	58	62
r5	252	250	247	212	76	76	78
r6	251	250	247	232	127	76	76

OpenCV output (`op03_brightness_opencv_output.csv`)

	c0	c1	c2	c3	c4	c5	c6
r0	252	247	191	78	70	61	81
r1	252	250	218	86	74	69	62
r2	253	251	225	123	64	62	58
r3	253	251	242	141	82	62	59
r4	253	250	244	191	70	58	62
r5	252	250	247	212	76	76	78
r6	251	250	247	232	127	76	76

Difference (manual - OpenCV) (`op03_brightness_difference.csv`)

	c0	c1	c2	c3	c4	c5	c6
r0	0	0	0	0	0	0	0
r1	0	0	0	0	0	0	0
r2	0	0	0	0	0	0	0
r3	0	0	0	0	0	0	0
r4	0	0	0	0	0	0	0
r5	0	0	0	0	0	0	0
r6	0	0	0	0	0	0	0

Maximum absolute difference: **0.0** (MATCH).

OP04 - Contrast adjustment (x1.25)

Input (`op04_contrast_input.csv`)

	c0	c1	c2	c3	c4	c5	c6
r0	212	207	151	38	30	21	41
r1	212	210	178	46	34	29	22
r2	213	211	185	83	24	22	18
r3	213	211	202	101	42	22	19
r4	213	210	204	151	30	18	22
r5	212	210	207	172	36	36	38
r6	211	210	207	192	87	36	36

Manual output (`op04_contrast_manual_output.csv`)

	c0	c1	c2	c3	c4	c5	c6
r0	255	255	189	48	38	26	51
r1	255	255	222	58	42	36	28
r2	255	255	231	104	30	28	22
r3	255	255	252	126	52	28	24
r4	255	255	255	189	38	22	28
r5	255	255	255	215	45	45	48

	c0	c1	c2	c3	c4	c5	c6
r6	255	255	255	240	109	45	45

OpenCV output (`op04_contrast_opencv_output.csv`)

	c0	c1	c2	c3	c4	c5	c6
r0	255	255	189	48	38	26	51
r1	255	255	222	58	42	36	28
r2	255	255	231	104	30	28	22
r3	255	255	252	126	52	28	24
r4	255	255	255	189	38	22	28
r5	255	255	255	215	45	45	48
r6	255	255	255	240	109	45	45

Difference (manual - OpenCV) (`op04_contrast_difference.csv`)

	c0	c1	c2	c3	c4	c5	c6
r0	0	0	0	0	0	0	0
r1	0	0	0	0	0	0	0
r2	0	0	0	0	0	0	0
r3	0	0	0	0	0	0	0
r4	0	0	0	0	0	0	0
r5	0	0	0	0	0	0	0
r6	0	0	0	0	0	0	0

Maximum absolute difference: **0.0** (MATCH).

OP05 - Binary thresholding (>127)

Input (`op05_threshold_input.csv`)

	c0	c1	c2	c3	c4	c5	c6
r0	212	207	151	38	30	21	41
r1	212	210	178	46	34	29	22
r2	213	211	185	83	24	22	18
r3	213	211	202	101	42	22	19
r4	213	210	204	151	30	18	22
r5	212	210	207	172	36	36	38
r6	211	210	207	192	87	36	36

Manual output (`op05_threshold_manual_output.csv`)

	c0	c1	c2	c3	c4	c5	c6
r0	255	255	255	0	0	0	0
r1	255	255	255	0	0	0	0
r2	255	255	255	0	0	0	0
r3	255	255	255	0	0	0	0
r4	255	255	255	255	0	0	0
r5	255	255	255	255	0	0	0
r6	255	255	255	255	0	0	0

OpenCV output (`op05_threshold_opencv_output.csv`)

	c0	c1	c2	c3	c4	c5	c6
r0	255	255	255	0	0	0	0
r1	255	255	255	0	0	0	0
r2	255	255	255	0	0	0	0
r3	255	255	255	0	0	0	0
r4	255	255	255	255	0	0	0
r5	255	255	255	255	0	0	0
r6	255	255	255	255	0	0	0

Difference (manual - OpenCV) (`op05_threshold_difference.csv`)

	c0	c1	c2	c3	c4	c5	c6
r0	0	0	0	0	0	0	0
r1	0	0	0	0	0	0	0
r2	0	0	0	0	0	0	0
r3	0	0	0	0	0	0	0

	c0	c1	c2	c3	c4	c5	c6
r4	0	0	0	0	0	0	0
r5	0	0	0	0	0	0	0
r6	0	0	0	0	0	0	0

Maximum absolute difference: **0.0** (MATCH).

OP06 - Horizontal flip

Input (`op06_flip_horizontal_input.csv`)

	c0	c1	c2	c3	c4	c5	c6
r0	212	207	151	38	30	21	41
r1	212	210	178	46	34	29	22
r2	213	211	185	83	24	22	18
r3	213	211	202	101	42	22	19
r4	213	210	204	151	30	18	22
r5	212	210	207	172	36	36	38
r6	211	210	207	192	87	36	36

Manual output (`op06_flip_horizontal_manual_output.csv`)

	c0	c1	c2	c3	c4	c5	c6
r0	41	21	30	38	151	207	212
r1	22	29	34	46	178	210	212
r2	18	22	24	83	185	211	213
r3	19	22	42	101	202	211	213
r4	22	18	30	151	204	210	213
r5	38	36	36	172	207	210	212
r6	36	36	87	192	207	210	211

OpenCV output (`op06_flip_horizontal_opencv_output.csv`)

	c0	c1	c2	c3	c4	c5	c6
r0	41	21	30	38	151	207	212
r1	22	29	34	46	178	210	212
r2	18	22	24	83	185	211	213
r3	19	22	42	101	202	211	213
r4	22	18	30	151	204	210	213
r5	38	36	36	172	207	210	212
r6	36	36	87	192	207	210	211

Difference (manual - OpenCV) (`op06_flip_horizontal_difference.csv`)

	c0	c1	c2	c3	c4	c5	c6
r0	0	0	0	0	0	0	0
r1	0	0	0	0	0	0	0
r2	0	0	0	0	0	0	0
r3	0	0	0	0	0	0	0
r4	0	0	0	0	0	0	0
r5	0	0	0	0	0	0	0
r6	0	0	0	0	0	0	0

Maximum absolute difference: **0.0** (MATCH).

OP07 - Mean filter (3x3)

Input (`op07_mean_input.csv`)

	c0	c1	c2	c3	c4	c5	c6
r0	212	207	151	38	30	21	41
r1	212	210	178	46	34	29	22
r2	213	211	185	83	24	22	18
r3	213	211	202	101	42	22	19
r4	213	210	204	151	30	18	22
r5	212	210	207	172	36	36	38
r6	211	210	207	192	87	36	36

Kernel (`op07_mean_kernel.csv`)

	c0	c1	c2
r0	0.1111	0.1111	0.1111
r1	0.1111	0.1111	0.1111
r2	0.1111	0.1111	0.1111

Manual output (`op07_mean_manual_output.csv`)

	c0	c1	c2	c3	c4
r0	197.667	145.444	85.4444	36.3333	26.7778
r1	203.889	158.556	99.4444	44.7778	25.7778
r2	206.889	173.111	113.556	54.7778	24.1111
r3	209.111	185.333	127.222	67.5556	29.2222
r4	209.333	195.889	142.889	84.2222	37.6667

OpenCV output (`op07_mean_opencv_output.csv`)

	c0	c1	c2	c3	c4
r0	197.667	145.444	85.4444	36.3333	26.7778
r1	203.889	158.556	99.4444	44.7778	25.7778
r2	206.889	173.111	113.556	54.7778	24.1111
r3	209.111	185.333	127.222	67.5556	29.2222
r4	209.333	195.889	142.889	84.2222	37.6667

Difference (manual - OpenCV) (`op07_mean_difference.csv`)

	c0	c1	c2	c3	c4
r0	-0	0	-0	0	0
r1	-0	-0	-0	0	0
r2	-0	-0	0	0	0
r3	-0	0	0	0	-0
r4	0	0	0	0	0

Worked calculations for three representative output cells

Output cell (0, 0)

Neighborhood values:

```
  212    207    151
 212    210    178
 213    211    185
```

`(1/9) x (212 + 207 + 151 + 212 + 210 + 178 + 213 + 211 + 185) = 1779/9 = 197.6667`

Manual result 197.6667, OpenCV result 197.6667.

Output cell (2, 2)

Neighborhood values:

```
  185     83     24
 202    101     42
 204    151     30
```

`(1/9) x (185 + 83 + 24 + 202 + 101 + 42 + 204 + 151 + 30) = 1022/9 = 113.5556`

Manual result 113.5556, OpenCV result 113.5556.

Output cell (4, 4)

Neighborhood values:

```
   30     18     22
  36     36     38
  87     36     36
```

`(1/9) x (30 + 18 + 22 + 36 + 36 + 38 + 87 + 36 + 36) = 339/9 = 37.6667`

Manual result 37.6667, OpenCV result 37.6667.

Maximum absolute difference: **0.0** (MATCH).

OP08 - Gaussian filter (3x3)

Input (`op08_gaussian_input.csv`)

	c0	c1	c2	c3	c4	c5	c6
r0	212	207	151	38	30	21	41
r1	212	210	178	46	34	29	22
r2	213	211	185	83	24	22	18
r3	213	211	202	101	42	22	19
r4	213	210	204	151	30	18	22
r5	212	210	207	172	36	36	38
r6	211	210	207	192	87	36	36

Kernel (`op08_gaussian_kernel.csv`)

	c0	c1	c2
r0	0.0625	0.125	0.0625
r1	0.125	0.25	0.125
r2	0.0625	0.125	0.0625

Manual output (`op08_gaussian_manual_output.csv`)

	c0	c1	c2	c3	c4
r0	201.062	152.188	77.5	34.875	26.6875
r1	205.438	166	93.75	41	24.4375
r2	208.188	179.062	112.688	49.75	24
r3	209.375	190.625	131.562	59.0625	26.6875
r4	209.562	198.562	149.25	74.4375	35.9375

OpenCV output (`op08_gaussian_opencv_output.csv`)

	c0	c1	c2	c3	c4
r0	201.062	152.188	77.5	34.875	26.6875
r1	205.438	166	93.75	41	24.4375
r2	208.188	179.062	112.688	49.75	24
r3	209.375	190.625	131.562	59.0625	26.6875
r4	209.562	198.562	149.25	74.4375	35.9375

Difference (manual - OpenCV) (`op08_gaussian_difference.csv`)

	c0	c1	c2	c3	c4
r0	0	0	0	0	0
r1	0	0	0	0	0
r2	0	0	0	0	0
r3	0	0	0	0	0
r4	0	0	0	0	0

Worked calculations for three representative output cells

Output cell (0, 0)

Neighborhood values:

```
  212     207     151
  212     210     178
  213     211     185
```

`(1/16) x [(1)(212) + (2)(207) + (1)(151) + (2)(212) + (4)(210) + (2)(178) + (1)(213) + (2)(211) + (1)(185)] = 3217/16 = 201.0625`

Manual result 201.0625, OpenCV result 201.0625.

Output cell (2, 2)

Neighborhood values:

```
  185      83      24
  202     101      42
  204     151      30
```

```
(1/16) x [(1)(185) + (2)(83) + (1)(24) + (2)(202) + (4)(101) + (2)(42) + (1)(204) + (2)(151) + (1)(30)] = 1803/16 = 112.6875
```

Manual result 112.6875, OpenCV result 112.6875.

Output cell (4, 4)

Neighborhood values:

```
  30     18     22
  36     36     38
  87     36     36
```

```
(1/16) x [(1)(30) + (2)(18) + (1)(22) + (2)(36) + (4)(36) + (2)(38) + (1)(87) + (2)(36) + (1)(36)] = 575/16 = 35.9375
```

Manual result 35.9375, OpenCV result 35.9375.

Maximum absolute difference: **0.0** (MATCH).

OP09 - Median filter (3x3)

Input (`op09_median_input.csv`)

	c0	c1	c2	c3	c4	c5	c6
r0	212	207	151	38	30	21	41
r1	212	210	178	46	34	29	22
r2	213	211	185	83	24	22	18
r3	213	211	202	101	42	22	19
r4	213	210	204	151	30	18	22
r5	212	210	207	172	36	36	38
r6	211	210	207	192	87	36	36

Manual output (`op09_median_manual_output.csv`)

	c0	c1	c2	c3	c4
r0	210	178	46	30	24
r1	211	185	83	34	22
r2	211	202	101	30	22
r3	210	204	151	36	30
r4	210	207	172	36	36

OpenCV output (`op09_median_opencv_output.csv`)

	c0	c1	c2	c3	c4
r0	210	178	46	30	24
r1	211	185	83	34	22
r2	211	202	101	30	22
r3	210	204	151	36	30
r4	210	207	172	36	36

Difference (manual - OpenCV) (`op09_median_difference.csv`)

	c0	c1	c2	c3	c4
r0	0	0	0	0	0
r1	0	0	0	0	0
r2	0	0	0	0	0
r3	0	0	0	0	0
r4	0	0	0	0	0

Worked calculations for three representative output cells

Output cell (0, 0)

Neighborhood values:

```
  212    207    151
  212    210    178
  213    211    185
```

```
sorted(151, 178, 185, 207, 210, 211, 212, 212, 213) -> middle value = 210
```

Manual result 210.0, OpenCV result 210.0.

Output cell (2, 2)

Neighborhood values:

```
 185    83    24
 202   101    42
 204   151    30
```

`sorted(24, 30, 42, 83, 101, 151, 185, 202, 204) -> middle value = 101`

Manual result 101.0, OpenCV result 101.0.

Output cell (4, 4)

Neighborhood values:

```
  30    18    22
  36    36    38
  87    36    36
```

`sorted(18, 22, 30, 36, 36, 36, 36, 38, 87) -> middle value = 36`

Manual result 36.0, OpenCV result 36.0.

Maximum absolute difference: **0.0** (MATCH).

OP10 - Sobel Gx

Input (`op10_sobel_gx_input.csv`)

	c0	c1	c2	c3	c4	c5	c6
r0	212	207	151	38	30	21	41
r1	212	210	178	46	34	29	22
r2	213	211	185	83	24	22	18
r3	213	211	202	101	42	22	19
r4	213	210	204	151	30	18	22
r5	212	210	207	172	36	36	38
r6	211	210	207	192	87	36	36

Kernel (`op10_sobel_gx_kernel.csv`)

	c0	c1	c2
r0	-1	0	1
r1	-2	0	2
r2	-1	0	1

Manual output (`op10_sobel_gx_manual_output.csv`)

	c0	c1	c2	c3	c4
r0	-157	-625	-570	-112	-19
r1	-101	-530	-626	-218	-47
r2	-59	-407	-655	-352	-60
r3	-34	-266	-679	-481	-37
r4	-23	-153	-636	-561	-55

OpenCV output (`op10_sobel_gx_opencv_output.csv`)

	c0	c1	c2	c3	c4
r0	-157	-625	-570	-112	-19
r1	-101	-530	-626	-218	-47
r2	-59	-407	-655	-352	-60
r3	-34	-266	-679	-481	-37
r4	-23	-153	-636	-561	-55

Difference (manual - OpenCV) (`op10_sobel_gx_difference.csv`)

	c0	c1	c2	c3	c4
r0	0	0	0	0	0
r1	0	0	0	0	0
r2	0	0	0	0	0
r3	0	0	0	0	0

	c0	c1	c2	c3	c4
r4	0	0	0	0	0

Worked calculations for three representative output cells

Output cell (0, 0)

Neighborhood values:

```
  212    207    151
 212    210    178
 213    211    185
```

`(−1)(212) + (0)(207) + (1)(151) + (−2)(212) + (0)(210) + (2)(178) + (−1)(213) + (0)(211) + (1)(185) = −157`

Manual result -157.0, OpenCV result -157.0.

Output cell (2, 2)

Neighborhood values:

```
  185     83     24
 202    101     42
 204    151     30
```

`(−1)(185) + (0)(83) + (1)(24) + (−2)(202) + (0)(101) + (2)(42) + (−1)(204) + (0)(151) + (1)(30) = −655`

Manual result -655.0, OpenCV result -655.0.

Output cell (4, 4)

Neighborhood values:

```
   30     18     22
  36     36     38
  87     36     36
```

`(−1)(30) + (0)(18) + (1)(22) + (−2)(36) + (0)(36) + (2)(38) + (−1)(87) + (0)(36) + (1)(36) = −55`

Manual result -55.0, OpenCV result -55.0.

Maximum absolute difference: **0.0** (MATCH).

OP11 - Sobel Gy

Input (`op11_sobel_gy_input.csv`)

	c0	c1	c2	c3	c4	c5	c6
r0	212	207	151	38	30	21	41
r1	212	210	178	46	34	29	22
r2	213	211	185	83	24	22	18
r3	213	211	202	101	42	22	19
r4	213	210	204	151	30	18	22
r5	212	210	207	172	36	36	38
r6	211	210	207	192	87	36	36

Kernel (`op11_sobel_gy_kernel.csv`)

	c0	c1	c2
r0	-1	-2	-1
r1	0	0	0
r2	1	2	1

Manual output (`op11_sobel_gy_manual_output.csv`)

	c0	c1	c2	c3	c4
r0	43	117	118	34	-27
r1	27	104	142	64	-9
r2	17	105	161	76	2
r3	2	80	141	73	41
r4	1	47	142	173	107

OpenCV output (`op11_sobel_gy_opencv_output.csv`)

	c0	c1	c2	c3	c4
r0	43	117	118	34	-27
r1	27	104	142	64	-9
r2	17	105	161	76	2
r3	2	80	141	73	41
r4	1	47	142	173	107

Difference (manual - OpenCV) (`op11_sobel_gy_difference.csv`)

	c0	c1	c2	c3	c4
r0	0	0	0	0	0
r1	0	0	0	0	0
r2	0	0	0	0	0
r3	0	0	0	0	0
r4	0	0	0	0	0

Worked calculations for three representative output cells

Output cell (0, 0)

Neighborhood values:

```
  212     207     151
 212     210     178
 213     211     185
```

`(−1)(212) + (−2)(207) + (−1)(151) + (0)(212) + (0)(210) + (0)(178) + (1)(213) + (2)(211) + (1)(185) = 43`

Manual result 43.0, OpenCV result 43.0.

Output cell (2, 2)

Neighborhood values:

```
  185      83      24
 202     101      42
 204     151      30
```

`(−1)(185) + (−2)(83) + (−1)(24) + (0)(202) + (0)(101) + (0)(42) + (1)(204) + (2)(151) + (1)(30) = 161`

Manual result 161.0, OpenCV result 161.0.

Output cell (4, 4)

Neighborhood values:

```
   30      18      22
  36      36      38
  87      36      36
```

`(−1)(30) + (−2)(18) + (−1)(22) + (0)(36) + (0)(36) + (0)(38) + (1)(87) + (2)(36) + (1)(36) = 107`

Manual result 107.0, OpenCV result 107.0.

Maximum absolute difference: **0.0** (MATCH).

OP12 - Sobel gradient magnitude

Input (`op12_sobel_magnitude_input.csv`)

	c0	c1	c2	c3	c4	c5	c6
r0	212	207	151	38	30	21	41
r1	212	210	178	46	34	29	22
r2	213	211	185	83	24	22	18
r3	213	211	202	101	42	22	19
r4	213	210	204	151	30	18	22
r5	212	210	207	172	36	36	38
r6	211	210	207	192	87	36	36

Manual output (`op12_sobel_magnitude_manual_output.csv`)

	c0	c1	c2	c3	c4
r0	162.782	635.857	582.086	117.047	33.0151
r1	104.547	540.107	641.903	227.2	47.8539
r2	61.4003	420.326	674.497	360.111	60.0333
r3	34.0588	277.77	693.485	486.508	55.2268
r4	23.0217	160.056	651.659	587.069	120.308

OpenCV output (`op12_sobel_magnitude_opencv_output.csv`)

	c0	c1	c2	c3	c4
r0	162.782	635.857	582.086	117.047	33.0151
r1	104.547	540.107	641.903	227.2	47.8539
r2	61.4003	420.326	674.497	360.111	60.0333
r3	34.0588	277.77	693.485	486.508	55.2268
r4	23.0217	160.056	651.659	587.069	120.308

Difference (manual - OpenCV) (`op12_sobel_magnitude_difference.csv`)

	c0	c1	c2	c3	c4
r0	0	0	0	0	0
r1	0	0	0	0	0
r2	0	0	0	0	0
r3	0	0	0	0	0
r4	0	0	0	0	0

Worked calculations for three representative output cells

Output cell (0, 0)

Gx = -157.0, Gy = 43.0

`sqrt((-157)^2 + (43)^2) = sqrt(26498) = 162.7821`

Manual result 162.7821, OpenCV result 162.7821.

Output cell (2, 2)

Gx = -655.0, Gy = 161.0

`sqrt((-655)^2 + (161)^2) = sqrt(454946) = 674.4968`

Manual result 674.4968, OpenCV result 674.4968.

Output cell (4, 4)

Gx = -55.0, Gy = 107.0

`sqrt((-55)^2 + (107)^2) = sqrt(14474) = 120.3079`

Manual result 120.3079, OpenCV result 120.3079.

Maximum absolute difference: **0.0** (MATCH).

OP13 - Erosion (3x3 ones)

Input (`op13_erosion_input.csv`)

	c0	c1	c2	c3	c4	c5	c6
r0	255	255	255	0	0	0	0
r1	255	255	255	0	0	0	0
r2	255	255	255	0	0	0	0
r3	255	255	255	0	0	0	0
r4	255	255	255	255	0	0	0
r5	255	255	255	255	0	0	0
r6	255	255	255	255	0	0	0

Kernel (`op13_erosion_kernel.csv`)

	c0	c1	c2
r0	1	1	1
r1	1	1	1
r2	1	1	1

Manual output (`op13_erosion_manual_output.csv`)

	c0	c1	c2	c3	c4
r0	255	0	0	0	0
r1	255	0	0	0	0
r2	255	0	0	0	0
r3	255	0	0	0	0
r4	255	255	0	0	0

OpenCV output (`op13_erosion_opencv_output.csv`)

	c0	c1	c2	c3	c4
r0	255	0	0	0	0
r1	255	0	0	0	0
r2	255	0	0	0	0
r3	255	0	0	0	0
r4	255	255	0	0	0

Difference (manual - OpenCV) (`op13_erosion_difference.csv`)

	c0	c1	c2	c3	c4
r0	0	0	0	0	0
r1	0	0	0	0	0
r2	0	0	0	0	0
r3	0	0	0	0	0
r4	0	0	0	0	0

Worked calculations for three representative output cells

Output cell (0, 0)

Neighborhood values:

```
  255     255     255
 255     255     255
 255     255     255
```

`min(255, 255, 255, 255, 255, 255, 255, 255, 255) = 255`

Manual result 255.0, OpenCV result 255.0.

Output cell (2, 2)

Neighborhood values:

```
  255       0       0
 255       0       0
 255     255       0
```

`min(255, 0, 0, 255, 0, 0, 255, 255, 0) = 0`

Manual result 0.0, OpenCV result 0.0.

Output cell (4, 4)

Neighborhood values:

```
    0       0       0
   0       0       0
   0       0       0
```

`min(0, 0, 0, 0, 0, 0, 0, 0, 0) = 0`

Manual result 0.0, OpenCV result 0.0.

Maximum absolute difference: **0.0** (MATCH).

OP14 - Dilation (3x3 ones)

Input (`op14_dilation_input.csv`)

	c0	c1	c2	c3	c4	c5	c6
r0	255	255	255	0	0	0	0
r1	255	255	255	0	0	0	0
r2	255	255	255	0	0	0	0
r3	255	255	255	0	0	0	0
r4	255	255	255	255	0	0	0
r5	255	255	255	255	0	0	0
r6	255	255	255	255	0	0	0

Kernel (`op14_dilation_kernel.csv`)

	c0	c1	c2
r0	1	1	1
r1	1	1	1
r2	1	1	1

Manual output (`op14_dilation_manual_output.csv`)

	c0	c1	c2	c3	c4
r0	255	255	255	0	0
r1	255	255	255	0	0
r2	255	255	255	255	0
r3	255	255	255	255	0
r4	255	255	255	255	0

OpenCV output (`op14_dilation_opencv_output.csv`)

	c0	c1	c2	c3	c4
r0	255	255	255	0	0
r1	255	255	255	0	0
r2	255	255	255	255	0
r3	255	255	255	255	0
r4	255	255	255	255	0

Difference (manual - OpenCV) (`op14_dilation_difference.csv`)

	c0	c1	c2	c3	c4
r0	0	0	0	0	0
r1	0	0	0	0	0
r2	0	0	0	0	0
r3	0	0	0	0	0
r4	0	0	0	0	0

Worked calculations for three representative output cells

Output cell (0, 0)

Neighborhood values:

```
  255    255    255
 255    255    255
 255    255    255
```

`max(255, 255, 255, 255, 255, 255, 255, 255, 255) = 255`

Manual result 255.0, OpenCV result 255.0.

Output cell (2, 2)

Neighborhood values:

```
  255      0      0
 255      0      0
 255    255      0
```

`max(255, 0, 0, 255, 0, 0, 255, 255, 0) = 255`

Manual result 255.0, OpenCV result 255.0.

Output cell (4, 4)

Neighborhood values:

```
  0     0     0
 0     0     0
 0     0     0
```

`max(0, 0, 0, 0, 0, 0, 0, 0, 0) = 0`

Manual result 0.0, OpenCV result 0.0.

Maximum absolute difference: **0.0** (MATCH).

6. Conclusion

Every operation applied through OpenCV corresponds to an arithmetic transformation of the underlying pixel matrix. The manual calculations reproduce the OpenCV results exactly, confirming that image processing is matrix processing.